

The critical little-Besov geometric mollification pipeline

Partial-result packet for `fa_banach_001`

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Status. Substantial partial progress toward the borderline $b_{3,\infty}^{2/3}$ conjecture in arXiv:2103.01723. The packet proves the sharp mollifier and commutator estimates, critical VMO recovery of the Stiefel constraint, all mollified-metric estimates needed by the source, existence and convergence of the weak second fundamental form, the weak Codazzi equations, and distributional vanishing of the canonical smooth Gauss determinants. It does *not* prove the endpoint compensated Jacobian/degree theorem required for developability.

Source and conjecture

Siran Li, Mohammad Reza Pakzad, and Armin Schikorra, *Fractional Sobolev isometric immersions of planar domains*, arXiv:2103.01723 (2021). In Remark 2.2, page 8 of the cached source PDF, the authors write:

“The estimates in Lemma 2.1 are not optimal and seem to characterize the spaces $b_{p,\infty}^s$, which are larger than $W^{s,p}$. We conjecture that results of the paper can still be achieved for the borderline space $b_{3,\infty}^{2/3}$ through the same approach.”

control as desired. □

Remark 2.2. *The estimates in Lemma 2.1 are not optimal and seem to characterize the spaces $b_{p,\infty}^s$ [44, Definition 2.1.3/1], which are larger than $W^{s,p}$. We conjecture that results of the paper can still be achieved for the borderline space $b_{3,\infty}^{2/3}$ through the same approach.*

Corollary 2.3 *Let $s \in (0, 1)$ and $n > 2$. If $f, a \in W^{s,p}(\Omega)$ where either Ω is smooth and*

The source proves C^1 regularity and developability for isometric immersions with $\nabla u \in W_{\text{loc}}^{2/3,3} = B_{3,3,\text{loc}}^{2/3}$. The conjectured space replaces the summable fine index 3 by the little endpoint fine index ∞ .

Little Besov notation

For $0 < s < 1$ and $1 \leq p < \infty$, use the difference norm

$$[f]_{B_{p,\infty}^s} = \sup_{0 < |h| \leq 1} |h|^{-s} \|f(\cdot - h) - f\|_{L^p}.$$

Write

$$\omega_f(r) = \sup_{0 < |h| \leq r} |h|^{-s} \|f(\cdot - h) - f\|_{L^p}.$$

The little space $b_{p,\infty}^s$ is the closure of smooth functions in $B_{p,\infty}^s$; equivalently, $\omega_f(r) \rightarrow 0$ as $r \downarrow 0$. On a smooth bounded domain all statements below are applied after a bounded extension and then localized away from the boundary.

Proof intuition

The word “little” supplies exactly the small factor missing from the large endpoint Besov space. Every translation by h is $o(|h|^s)$ in L^p , uniformly for smaller translations. Averaging this identity gives all of the source’s $o(\varepsilon^{s-k})$ mollifier estimates and, after applying it twice, the quadratic commutator estimates.

At the critical index $sp = n$, the same translation modulus controls the mean oscillation on *every* ball, uniformly in its center. Hence a little-Besov map taking values almost everywhere in a compact manifold has mollifications uniformly approaching that manifold. For an isometric gradient the manifold is the Stiefel manifold $O(2, 3)$; this recovers the uniform nondegeneracy that the geometric proof needs. Standard positive-sum Besov paraproduct estimates then define the weak second fundamental form. Codazzi and Gauss identities for the smooth mollifications survive because their error terms contain precisely the vanishing factors obtained above.

Endpoint analytic calculus

Fix a standard compactly supported mollifier φ and write $f_\varepsilon = f * \varphi_\varepsilon$.

Lemma 1 (Little-Besov mollifier estimates). *Let $f, g \in b_{p,\infty}^s(\mathbb{R}^n)$, $0 < s < 1$, and $p \geq 2$. For every integer $k \geq 1$,*

$$\begin{aligned} \|f_\varepsilon - f\|_{L^p} &= o(\varepsilon^s), \\ \|\nabla^k f_\varepsilon\|_{L^p} &= o(\varepsilon^{s-k}), \\ \|\nabla^k(f_\varepsilon g_\varepsilon - (fg)_\varepsilon)\|_{L^{p/2}} &= o(\varepsilon^{2s-k}) \quad (k \geq 0). \end{aligned}$$

Moreover,

$$f_\varepsilon g_\varepsilon - (fg)_\varepsilon \longrightarrow 0 \quad \text{in } W^{2s,p/2}.$$

Proof. Translation averaging gives

$$\|f_\varepsilon - f\|_p \leq \int \varphi_\varepsilon(h) \|f(\cdot - h) - f\|_p dh \leq C\varepsilon^s \omega_f(\varepsilon).$$

For $k \geq 1$, use $\int \nabla^k \varphi_\varepsilon = 0$ to obtain

$$\|\nabla^k f_\varepsilon\|_p \leq \int |\nabla^k \varphi_\varepsilon(h)| \|f(\cdot - h) - f\|_p dh \leq C\varepsilon^{s-k} \omega_f(\varepsilon).$$

Both right-hand sides have the claimed little- o factor.

For $k = 0$ use the exact identity

$$f_\varepsilon g_\varepsilon - (fg)_\varepsilon = (f_\varepsilon - f)(g_\varepsilon - g) - \int \varphi_\varepsilon(h) \delta_h f \delta_h g dh,$$

where $\delta_h f = f(\cdot - h) - f$. Hölder’s inequality bounds both terms by $C\varepsilon^{2s} \omega_f(\varepsilon) \omega_g(\varepsilon)$. For $k \geq 1$, differentiating and using $\int \nabla^k \varphi_\varepsilon = 0$ gives a finite sum of terms

$$\nabla^j f_\varepsilon \nabla^{k-j} g_\varepsilon, quad(f_\varepsilon - f) \nabla^k g_\varepsilon, quad(g_\varepsilon - g) \nabla^k f_\varepsilon, quad \int \nabla^k \varphi_\varepsilon(h) \delta_h f \delta_h g dh.$$

The preceding estimates and Hölder give $o(\varepsilon^{2s-k})$ for each one.

If $s \leq 1/2$, interpolate the $k = 0, 1$ bounds with parameter $2s$. If $s > 1/2$, interpolate the $k = 1, 2$ bounds for the gradient with parameter $2s - 1$, and combine with the $k = 0$ bound. The powers of ε cancel and the little factors remain, proving strong convergence in $W^{2s,p/2}$. $\square$

Lemma 2 (Critical uniform VMO and manifold recovery). *Assume $sp = n$, let $\Omega \subset \mathbb{R}^n$ be a smooth bounded domain, and let $F \in b_{p,\infty}^s(\Omega; \mathbb{R}^N)$. For every ball $B_{2r}(x) \Subset \Omega$,*

$$\left(\oint_{B_r(x)} |F - F_{B_r(x)}|^p \right)^{1/p} \leq C\omega_F(2r).$$

If in addition $F(x) \in K$ almost everywhere for a compact set $K \subset \mathbb{R}^N$, then, for every $\Omega' \Subset \Omega$,

$$\sup_{x \in \Omega'} \text{dist}(F_\varepsilon(x), K) \longrightarrow 0.$$

Proof. By Jensen and the change of variables $h = y - x$,

$$\begin{aligned} \oint_{B_r} |F - F_{B_r}|^p &\leq |B_r|^{-2} \int_{B_r} \int_{B_r} |F(y) - F(z)|^p dy dz \\ &\leq Cr^{-2n} \int_{|h| \leq 2r} \|\delta_h F\|_p^p dh \\ &\leq C\omega_F(2r)^p r^{-2n} \int_{|h| \leq 2r} |h|^{sp} dh \leq C\omega_F(2r)^p, \end{aligned}$$

because $sp = n$. For manifold recovery, compare $F_\varepsilon(x)$ with the unweighted average $F_{B_\varepsilon(x)}$. The boundedness of the mollifier gives

$$|F_\varepsilon(x) - F_{B_\varepsilon(x)}| \leq C \left(\oint_{B_\varepsilon(x)} |F - F_{B_\varepsilon(x)}|^p \right)^{1/p}.$$

Since $F(y) \in K$ almost everywhere, $\text{dist}(F_{B_\varepsilon(x)}, K)$ is bounded by the same mean oscillation. Lemma's first assertion and $\omega_F(2\varepsilon) \rightarrow 0$ finish the proof. $\square$

Geometric endpoint theorem

Let

$$V_2(\mathbb{R}^3) = \{A \in \mathbb{R}^{3 \times 2} : A^T A = I_2\}$$

be the Stiefel manifold.

Theorem 3 (The geometric pipeline through Gauss–Codazzi). *Let $\Omega \subset \mathbb{R}^2$ be open and let $u \in W_{\text{loc}}^{1,\infty}(\Omega; \mathbb{R}^3)$ satisfy*

$$F := \nabla u \in b_{3,\infty,\text{loc}}^{2/3}, \quad F^T F = I_2 \quad \text{a.e.}$$

For $\Omega' \Subset \Omega$, mollify inside a fixed intermediate domain and put

$$g^\varepsilon = F_\varepsilon^T F_\varepsilon, \quad n^\varepsilon = \frac{\partial_1 u_\varepsilon \wedge \partial_2 u_\varepsilon}{|\partial_1 u_\varepsilon \wedge \partial_2 u_\varepsilon|}, \quad \Pi_{ij}^\varepsilon = \partial_{ij} u_\varepsilon \cdot n^\varepsilon.$$

Then:

1. $\sup_{\Omega'} \text{dist}(F_\varepsilon, V_2(\mathbb{R}^3)) \rightarrow 0$. Consequently $g^\varepsilon \rightarrow I_2$ uniformly and u_ε is an immersion for all small ε .

2. The metric satisfies

$$\begin{aligned} \|g^\varepsilon - I_2\|_{L^3} &= o(\varepsilon^{2/3}), & \|\nabla g^\varepsilon\|_{L^3} &= o(\varepsilon^{-1/3}), \\ \|\nabla^k(g^\varepsilon - I_2)\|_{L^{3/2}} &= o(\varepsilon^{4/3-k}) & (k \geq 0), \end{aligned}$$

and $g^\varepsilon - I_2 \rightarrow 0$ in $W^{4/3, 3/2}$.

3. The forms Π^ε converge in distributions to a canonical weak second fundamental form $\Pi \in b_{3,\infty,\text{loc}}^{-1/3}$. It satisfies $\text{curl } \Pi = 0$. Hence, on every simply connected $U \Subset \Omega$, there is $q \in b_{3,\infty,\text{loc}}^{2/3}(U; \mathbb{R}^2)$ with $\nabla q = \Pi$.

4. The canonical smooth Gauss determinants vanish:

$$\det \Pi^\varepsilon \rightarrow 0 \quad \text{in } \mathcal{D}'(\Omega).$$

Proof. Apply Lemma 2 with $K = V_2(\mathbb{R}^3)$ to obtain (1). Uniform closeness to this compact manifold also gives uniform boundedness of F_ε and uniform invertibility of g^ε . Since F itself is bounded, Lemma 1 gives

$$\|g^\varepsilon - I_2\|_3 \leq C\|F_\varepsilon - F\|_3 = o(\varepsilon^{2/3}), \quad \|\nabla g^\varepsilon\|_3 \leq C\|\nabla F_\varepsilon\|_3 = o(\varepsilon^{-1/3}).$$

Also

$$g^\varepsilon - I_2 = F_\varepsilon^T F_\varepsilon - (F^T F)_\varepsilon,$$

so the remaining claims in (2) are exactly Lemma 1, applied componentwise.

For completeness, the Besov product used next is standard: if $1/2 < s < 1$, $p = 2/s$, and $a, b \in b_{p,\infty}^s \cap L^\infty$, then

$$a \partial_j b \in b_{p,\infty}^{s-1} \quad (1)$$

continuously. A Bony decomposition proves this directly. The low-high terms use the L^∞ factors; the resonant high-high term lies in $B_{p/2,\infty}^{2s-1}$ because $2s-1 > 0$, and the critical Sobolev embedding $B_{p/2,\infty}^{2s-1} \hookrightarrow B_{p,\infty}^{s-1}$ finishes the estimate. Density makes the map continuous on the little subspaces.

The smooth normal map on a neighborhood of $V_2(\mathbb{R}^3)$, the approximation property in the little space, and (1) show that $n^\varepsilon \rightarrow n = \partial_1 u \wedge \partial_2 u$ in $b_{3,\infty,\text{loc}}^{2/3}$ and that $\Pi^\varepsilon = n^\varepsilon \cdot \nabla F_\varepsilon$ converges in $b_{3,\infty,\text{loc}}^{-1/3}$ to a canonical distribution Π .

For the smooth immersion u_ε , the Codazzi equations give

$$\partial_2 \Pi_{i1}^\varepsilon - \partial_1 \Pi_{i2}^\varepsilon = \Pi_{\ell 1}^\varepsilon \Gamma_{i2}^{\ell,\varepsilon} - \Pi_{\ell 2}^\varepsilon \Gamma_{i1}^{\ell,\varepsilon}.$$

The established estimates imply

$$\|\Pi^\varepsilon\|_3 = o(\varepsilon^{-1/3}), \quad \|\Gamma^\varepsilon\|_{3/2} = o(\varepsilon^{1/3}).$$

Hölder therefore makes the Codazzi right-hand side $o(1)$ in L^1 . Passing to the distributional limit gives $\text{curl } \Pi = 0$. Local Hodge theory, or equivalently order-zero Fourier multipliers after a cutoff, is bounded on Besov spaces and produces the asserted potential q .

Finally, differentiating the metric estimates gives

$$\|\Gamma^\varepsilon\|_{3/2} = o(\varepsilon^{1/3}), \quad \|\nabla \Gamma^\varepsilon\|_{3/2} = o(\varepsilon^{-2/3}).$$

Interpolation yields $\Gamma^\varepsilon \rightarrow 0$ in $W^{1/3,3/2} \hookrightarrow L^2$. The smooth Gauss identity, in the coordinate form used by the source, reads

$$\det \Pi^\varepsilon = -\operatorname{curl}^T \operatorname{curl} g^\varepsilon + O(|\Gamma^\varepsilon|^2).$$

After pairing with a test function and integrating once, the first term is bounded by $\|\nabla g^\varepsilon\|_{3/2} \|\nabla \phi\|_3 = o(\varepsilon^{1/3})$; the second is bounded by $\|\Gamma^\varepsilon\|_2^2 \|\phi\|_\infty = o(1)$. This proves (4). $\square$

What remains for the full conjecture

The theorem reaches the exact point where the fine Besov index matters in a new way. To invoke the source’s planar developability theorem one needs an intrinsic distributional Jacobian $\operatorname{Jac}(q)$, stability of that Jacobian under the relevant approximations, continuity or an adequate representative of q , and the topological degree formula. In $W^{2/3,3} = B_{3,3}^{2/3}$ these are supported by summation over scales. The condition defining $b_{3,\infty}^{2/3}$ says only that the individual dyadic coefficients tend to zero; it does not make their cubes or quadratic interactions summable. Thus the limit $\det \Pi^\varepsilon \rightarrow 0$ along the canonical geometric approximation does not, by itself, prove that every admissible approximation defines the same $\operatorname{Jac}(q)$.

Nor does critical $b_{3,\infty}^{2/3}$ alone embed into continuous functions. The missing planar theorem must exploit both $\operatorname{curl} q = 0$ and the vanishing Gauss/Jacobian structure. Resolving that compensated endpoint is a credible route to the full conjecture, but it is not contained in the present proof.

Verification, novelty search, and review recommendation

Every exponent in Theorem 3 is scale invariant at $(s, p, n) = (2/3, 3, 2)$. The two critical products are explicitly checked: $L^3 \cdot L^{3/2} \subset L^1$ for Codazzi, and $W^{1/3,3/2} \hookrightarrow L^2$ for the quadratic Christoffel term. The conical singularity discussed by the source lies in the large space $B_{3,\infty}^{2/3}$ with nonvanishing scale modulus, but not in the little space, so it does not contradict the result.

The run indexes contained no duplicate for arXiv:2103.01723 or this Besov endpoint. Bounded arXiv searches for the exact $b_{3,\infty}^{2/3}$ notation, the title, and “Besov developability” returned the source paper but no later resolution. Validity confidence for the proved analytic/geometric pipeline is high; novelty confidence is moderate because Remark 2.2 predicts the mollifier part, though it does not state the uniform manifold, weak Codazzi, or Gauss conclusions proved here.

Human-review recommendation: promote as a substantial partial. The reviewer should focus on the localization of the paraproduct estimate (1) and the passage from curl-free Besov distributions to the local potential. Do not label this a full solution unless the endpoint compensated-Jacobian and planar degree/developability step is separately proved.

References

- [1] S. Li, M. R. Pakzad, and A. Schikorra, *Fractional Sobolev isometric immersions of planar domains*, arXiv:2103.01723, 2021.
- [2] T. Runst and W. Sickel, *Sobolev Spaces of Fractional Order, Nemytskij Operators, and Nonlinear Partial Differential Equations*, de Gruyter, 1996.